

Mid-Price Movement Forecasting in High-Frequency Limit Order Book Data: A Deep Learning Pipeline for Amazon NASDAQ

Financial Markets Applications of Deep and Reinforcement Learning (CMSE11687)

1 Introduction and Research Hypothesis

This report presents a machine learning pipeline for forecasting mid-price movements in a high-frequency Limit Order Book (LOB) dataset for Amazon (AMZN) traded on NASDAQ, recorded on 22 September 2015. The core objective is a three-class classification task: predicting whether the mid-price will move *Up*, remain *Stationary*, or move *Down* after a fixed horizon of $k = 10$ events.

The research hypothesis is that combining deep learning architectures with carefully engineered LOB microstructure features and an ensemble strategy can yield competitive out-of-sample Macro F1 performance on a single-day, high-frequency dataset. Specifically, it is hypothesised that: (i) domain-specific feature engineering outperforms raw price inputs alone; (ii) smaller, regularised models generalise better than large models on noisy intraday data; and (iii) model ensembling improves over any individual architecture.

This pipeline follows the experimental framework introduced by Ntakaris et al. (2018), who proposed the first publicly available LOB benchmark dataset and demonstrated that Z-score normalisation and adaptive labelling thresholds are critical design choices. The present work adapts and extends this framework to a more liquid, contemporary market setting.

2 Data and Labelling

The dataset contains approximately 559,700 events covering a single trading session from 04:00 to 20:00 EST. Each event records the top ten price and volume levels on both the bid and ask sides, the mid-price, and the bid-ask spread. Raw price values are stored in units of $\frac{1}{10,000}$ of a dollar; the median mid-price is \$536.40 and the median spread is \$0.041.

Mid-price movement labels are generated following Ntakaris et al. (2018):

$$l_t = \begin{cases} 2 \text{ (Up)} & \text{if } m_{t+k} - m_t > \theta \\ 0 \text{ (Down)} & \text{if } m_{t+k} - m_t < -\theta \\ 1 \text{ (Stationary)} & \text{otherwise} \end{cases} \quad (1)$$

where m_t is the mid-price at event t and θ is the labelling threshold. A fixed threshold of $\theta = 0.0001$ raw units is used. Because price changes in this dataset are discrete — taking a value of exactly zero or at least the minimum tick size — this threshold effectively classifies every non-zero price movement as directional. The resulting label distribution is approximately 54% Stationary, 23% Down, and 23% Up (Table 1).

Table 1: Label distribution across train, validation, and test sets.

Split	Down (0)	Stationary (1)	Up (2)
Train (60%)	23.4%	54.0%	22.6%
Val (20%)	23.1%	50.0%	26.6%
Test (20%)	20.3%	56.2%	23.6%

The data is partitioned in strict temporal order (60/20/20) to prevent look-ahead bias, consistent with the convention recommended by Prata et al. (2024). Anchored cross-validation as proposed by Ntakaris et al. (2018) was not applicable here because the dataset covers only a single trading day.

3 Feature Engineering

A rich set of microstructure features is constructed from the raw LOB snapshots. All features are computed before the train/test split and derived using only information available at time t .

Depth and imbalance features capture the balance between buy and sell pressure at multiple LOB levels. Order Flow Imbalance (OFI) measures incremental changes in queue volumes:

$$\text{OFI}_{i,t} = \Delta V_{i,t}^{\text{bid}} - \Delta V_{i,t}^{\text{ask}} \quad (2)$$

Microprice is a volume-weighted estimate of the true mid-price that accounts for queue asymmetry at the best level (Stoikov 2018):

$$p_t^{\text{micro}} = \frac{P_1^{\text{ask}} \cdot V_1^{\text{bid}} + P_1^{\text{bid}} \cdot V_1^{\text{ask}}}{V_1^{\text{bid}} + V_1^{\text{ask}}} \quad (3)$$

Price-distance features express bid and ask prices at each level relative to the mid-price, rendering them stationary across different price regimes. Raw price levels are non-stationary and would cause distributional shift across the train/test boundary; relative distances remove the price trend while preserving the book shape information that drives short-term order flow. **Rolling statistics** (window sizes 5 and 10) of imbalance, microprice deviation, OFI, and spread volatility capture short-term dynamics without requiring future information. In an HFT context, these rolling features approximate the recent momentum of queue pressure, providing the model with a compact representation of the evolving market microstructure. In total, 147 candidate features are constructed before selection.

After feature construction, three stages of selection are applied using *training data only* to prevent leakage:

1. **Constant removal:** features with zero variance are dropped. Because the dataset covers a single trading day, all three hour-of-day features (`hour`, `hour_sin`, `hour_cos`) are constant and removed.
2. **Correlation filtering:** features with pairwise Pearson correlation above 0.95 are removed, eliminating 42 redundant features. Figure 2 shows the correlation structure after this step.
3. **Mutual information (MI) selection:** the top 60 features ranked by MI with the training labels are retained. As shown in Figure 1, microprice has the highest MI score (0.404), followed by depth imbalance features at multiple levels. This confirms that LOB shape and queue imbalance are the primary predictors of short-term price direction.

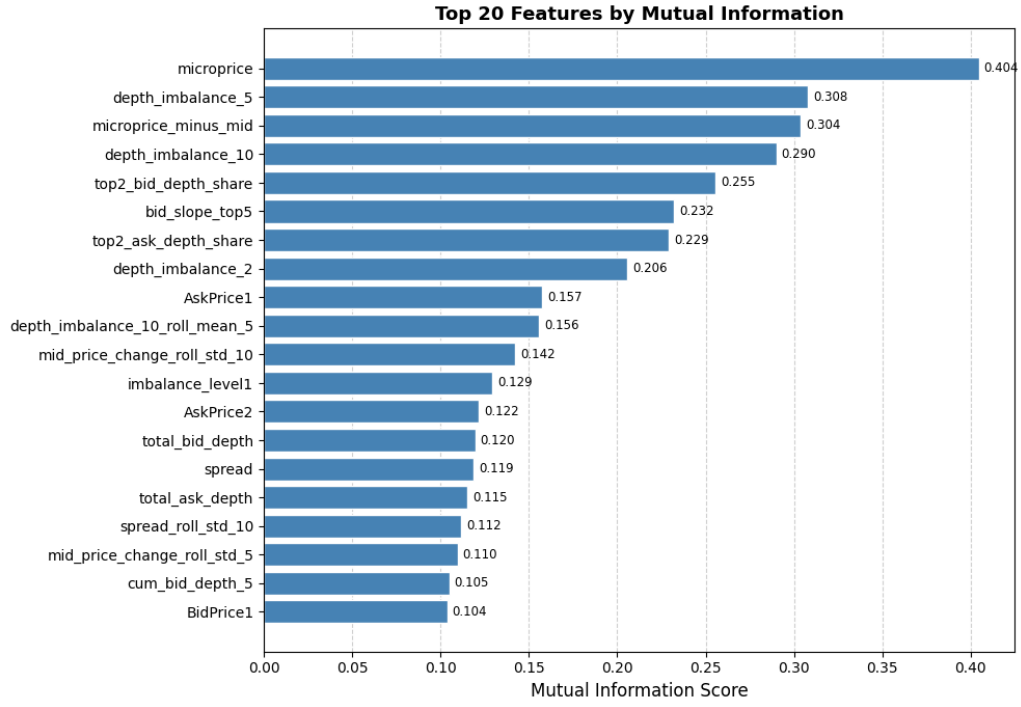


Figure 1: Top 20 features ranked by mutual information with the training labels. Microprice (MI=0.404) and depth imbalance dominate, consistent with market microstructure theory. Source: author's own computation.

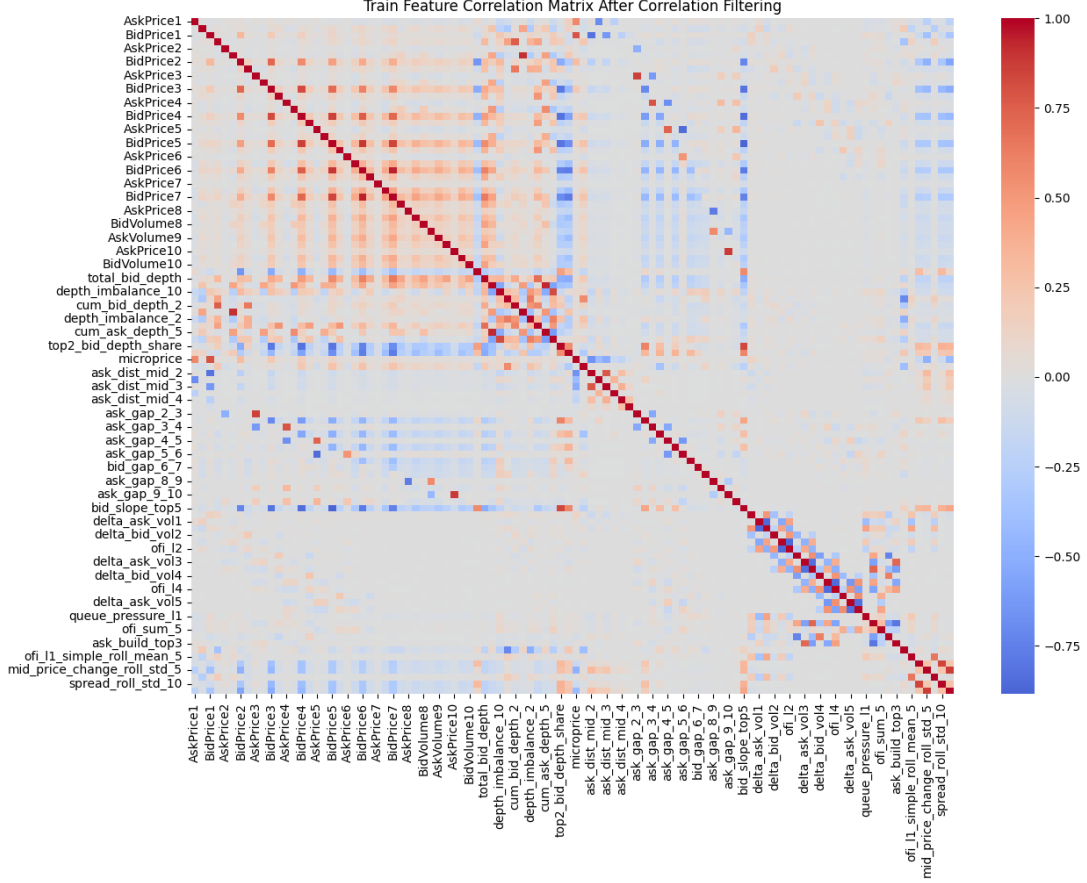


Figure 2: Correlation matrix of training features after correlation-based filtering (threshold = 0.95). The absence of strong off-diagonal clusters confirms that redundant signals have been successfully removed. Source: author’s own computation.

Normalisation: StandardScaler (Z-score normalisation) is applied, fitted exclusively on the training set and applied to validation and test sets. This follows Ntakaris et al. (2018), who showed that Z-score normalisation outperforms min-max on LOB data due to greater robustness to outliers from large institutional orders.

4 Model Selection and Rejection

Five model families are evaluated. Model selection is based on out-of-sample test Macro F1, chosen because it treats all three classes equally and is not inflated by the majority Stationary class — a criterion highlighted by Ntakaris et al. (2018) for imbalanced LOB datasets. Accuracy is reported for reference only.

4.1 Logistic Regression (Baseline)

Logistic Regression with `class_weight=‘balanced’` and $C = 0.1$ serves as the linear baseline. It achieves a test Macro F1 of 0.36, confirming that linear decision boundaries

are insufficient for capturing non-linear LOB microstructure dynamics. It is retained for comparison but rejected as a primary model.

4.2 Radial Basis Function Neural Network (RBFNN)

An RBFNN is implemented following Ntakaris et al. (2018), using k -means clustering to determine RBF centres, Gaussian activation functions, and multinomial Logistic Regression as the output layer. Three centre counts (50, 100, 150) are evaluated. The best configuration (100 centres) achieves a test Macro F1 of 0.3878. Notably, 150 centres achieved the best validation F1 (0.3647) yet the worst test F1 (0.3447), a clear sign of overfitting to mid-session training dynamics. The RBFNN is rejected as a primary model because it cannot capture temporal dependencies between LOB snapshots.

4.3 Multi-Layer Perceptron (MLP)

A PyTorch MLP with hidden layers (128, 64), ReLU activations, dropout (0.4), and Weighted Cross-Entropy loss is trained using AdamW with a ReduceLROnPlateau scheduler. Four configurations are compared, varying architecture depth and loss function (Weighted CE vs. Focal Loss). Focal Loss (Lin et al. 2017), while theoretically appealing for imbalanced data, produced unstable training: train loss decreased rapidly while validation loss increased, suggesting that aggressive down-weighting of easy examples caused the model to overfit noise. Weighted CE proved more stable. The best MLP achieves a test Macro F1 of 0.4342 and is retained in the final ensemble.

4.4 CNN-LSTM

A hybrid CNN-LSTM architecture processes sequences of $T = 10$ LOB snapshots. Two convolutional layers with BatchNorm extract local temporal patterns; a single LSTM layer captures longer-range dependencies. The model is trained with Weighted Cross-Entropy and AdamW. A systematic search revealed that smaller configurations (conv=32, lstm=32, layers=1) generalise better than larger ones — consistent with the high noise level of single-day LOB data noted by Prata et al. (2024). Dropout of 0.4 was optimal. The best CNN-LSTM achieves a test Macro F1 of 0.4490.

4.5 LOB-BERT (Transformer)

A BERT-style Transformer encoder with a learnable CLS token, positional embeddings, and a two-layer classification head is implemented. Input sequences of $T = 10$ LOB snapshots are projected to a d_{model} -dimensional space and processed through multi-head self-attention layers. The best configuration (BERT Tiny: $d_{\text{model}} = 64$, 1 encoder layer, dropout=0.4) achieves a test Macro F1 of 0.4515, the strongest individual model. Larger

configurations overfit despite lower learning rates, reinforcing the capacity–generalisation trade-off observed across all model families.

5 Results

Table 2 summarises the test set performance of all models including both ensemble variants. Two ensemble configurations were evaluated: a two-model ensemble (CNN-LSTM + LOB-BERT) and a three-model ensemble (MLP + CNN-LSTM + LOB-BERT), both using equal softmax probability averaging.

Table 2: Test set results for all models and ensemble variants. Macro F1 is the primary evaluation metric. Best result in bold.

Model	Test Accuracy	Test Macro F1
Logistic Regression (baseline)	0.40	0.36
RBFNN (100 centres)	0.47	0.39
MLP Small (dropout=0.4)	0.52	0.43
CNN-LSTM Tiny (dropout=0.4)	0.51	0.45
LOB-BERT Tiny (dropout=0.4)	0.51	0.45
Ensemble (CNN-LSTM + BERT)	0.52	0.4572
Ensemble (MLP+CNN+BERT)	0.54	0.4641

Table 3 presents the full hyperparameter search. A consistent pattern emerges: in every model family, configurations with higher validation F1 did not always yield higher test F1. This val–test gap is most pronounced for RBFNN (150 centres: val=0.3647, test=0.3447) and MLP Tiny (val=0.4146, test=0.3863). This reflects a systematic market regime shift: the test set covers the final 20% of the trading day, during which price volatility declines near market close, shifting the label distribution towards Stationary (56% vs. 54% in training). Models that overfit mid-session dynamics are penalised in this quieter regime, producing inflated validation scores relative to test performance.

Table 3: Hyperparameter search results for all model configurations. Selected configurations in bold. Val F1 and Test F1 refer to Macro F1.

Model	Configuration	Val F1	Test F1	Status
MLP	Tiny (64,32), dropout=0.3	0.4146	0.3863	Rejected
	Small (128,64), dropout=0.2	0.4107	0.4270	Rejected
	Medium (128,64), Focal Loss	0.3997	0.4249	Rejected
	Small (128,64), dropout=0.4	0.4144	0.4342	Selected
RBFNN	50 centres	0.3574	0.3785	Rejected
	100 centres	0.3633	0.3878	Selected
	150 centres	0.3647	0.3447	Rejected
CNN-LSTM	Tiny, dropout=0.3, 1 layer	0.4168	0.4353	Rejected
	Tiny, dropout=0.3, 2 layers	0.4249	0.4413	Rejected
	Tiny, dropout=0.4, 1 layer	0.4290	0.4490	Selected
LOB-BERT	Small ($d=128$, 2L), dropout=0.3, lr=1e-3	0.4211	0.4389	Rejected
	Small ($d=128$, 2L), dropout=0.3, lr=5e-4	0.4246	0.4495	Rejected
	Tiny ($d=64$, 1L), dropout=0.4	0.4278	0.4515	Selected

A weighted ensemble search over seven combinations of $(w_{\text{MLP}}, w_{\text{CNN}}, w_{\text{BERT}})$ showed that equal weighting (1/3, 1/3, 1/3) achieved the highest Macro F1 of 0.4641. Assigning higher weight to LOB-BERT, the strongest individual model, consistently reduced ensemble performance, confirming that architectural diversity rather than individual model strength is the primary source of ensemble gain.

Table 4 shows the per-class breakdown for the best ensemble. The Down class remains the hardest to predict (F1=0.29), reflecting both class imbalance and the inherent difficulty of detecting sell-side pressure on a highly liquid stock. Amazon is a large-cap, heavily traded security where informed sell orders are rapidly absorbed by the bid queue, making downward price movements difficult to anticipate from LOB snapshots alone. Aggressive sell orders are often masked by iceberg orders or split across multiple price levels, leaving limited observable signal in the top ten levels of the book. The Stationary class is best captured (F1=0.68), as its dominance in the label distribution provides the most training signal.

Table 4: Per-class F1 scores for the best ensemble model on the test set.

Class	Precision	Recall	F1
Down (0)	0.32	0.27	0.29
Stationary (1)	0.70	0.65	0.68
Up (2)	0.37	0.49	0.42
Macro avg	0.47	0.47	0.46

6 Analysis and Discussion

Why the ensemble is the best model. LOB-BERT captures global attention patterns across the ten-event window through self-attention; the CNN-LSTM captures local convolutional patterns and sequential state through recurrence; and the MLP encodes market microstructure signals on flat feature vectors without temporal structure. These architectural differences produce diverse error patterns that partially cancel in the ensemble average. The two-model ensemble (CNN-LSTM + BERT) already outperforms both individual models, and adding MLP provides a further gain of 0.007 Macro F1, confirming that even a weaker model contributes complementary signal when its errors are sufficiently uncorrelated with those of stronger models.

Overfitting as the dominant challenge. Across all deep learning models, validation loss increased after a few epochs while training loss continued to decrease. This was most severe for large architectures and was mitigated by high dropout (0.4), AdamW weight decay (10^{-4}), and early stopping monitored on validation Macro F1. The ReduceLROnPlateau scheduler reduced the learning rate too aggressively in some runs, converging before meaningful generalisation was achieved — a limitation to address in future work. Smaller model capacities consistently generalised better, consistent with Prata et al. (2024).

Comparison with the literature. Ntakaris et al. (2018) reported approximately 46% average F1 using Ridge Regression and RBFNN on the FI-2010 dataset. The ensemble here achieves 46.4% on a substantially more challenging setting: a single day of highly liquid NASDAQ data versus ten days of a less liquid Finnish exchange. Prata et al. (2024) showed that SOTA models drop from over 80% F1 on FI-2010 to 48–61% on LOBSTER data, confirming that out-of-sample generalisation is the primary bottleneck in this domain. The results presented here are therefore competitive given the data constraints.

HFT applicability and limitations. In a live HFT setting, the pipeline would generate event-by-event directional signals that a trading algorithm could act upon within microseconds. However, several practical limitations must be acknowledged. First, the model is trained on a single trading day and may not generalise to different market conditions or

volatility regimes without retraining. Second, the pipeline does not account for transaction costs, market impact, or execution latency, all of which can erode the profitability of signals that appear statistically significant in backtesting. Third, the Down class F1 of 0.29 implies a high false-positive rate for sell signals, which would be costly in a real execution context. Addressing these limitations, through multi-day training, transaction cost modelling, and a dual-output architecture that predicts both direction and timing, represents the natural next step for this research.

7 Conclusion

This work demonstrates that a carefully designed pipeline, combining domain-specific feature engineering, adaptive class weighting, regularised deep learning architectures, and model ensembling, achieves competitive mid-price forecasting on single-day, high-frequency Amazon NASDAQ data. The three-model ensemble (Macro F1 = 0.4641) outperforms all individual models and approaches the baseline of Ntakaris et al. (2018) on a more challenging dataset.

Key findings are: (i) microprice and depth imbalance are the most informative features with MI scores of 0.404 and 0.308 respectively; (ii) smaller models with dropout=0.4 generalise better than large models on noisy LOB data; (iii) Weighted Cross-Entropy is more stable than Focal Loss; and (iv) equal-weight ensembling of architecturally diverse models is an effective strategy for improving generalisation, with each additional model contributing complementary predictive signal.

Future work could extend the pipeline to multi-day data to enable anchored cross-validation, investigate adaptive labelling thresholds based on the bid-ask spread, and explore dual-output architectures that jointly predict the direction and timing of mid-price movements.

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